

S2CHANGE: Desenvolvimento de mapas de perdas recentes de floresta e mato em Portugal derivados de imagens de satélite

Contrato N.º 3044 de cooperação entre a Direção Geral do Território e o Instituto Superior de Agronomia

Entregável 4.2. Demonstrador prático: mapas nacionais a identificar o agente causador das perdas recentes de floresta e mato superiores a 0.5 ha produzidas na tarefa 3.

(O relatório é redigido em Inglês)

Contents

Introduction	1
Data	2
Dates and tiles.....	2
Vegetation loss map features and attributes	4

Introduction

ISA_DGT_INCD_DL_pipeline is the computation pipeline developed at the INCD HPC cluster, operated by LIP, to process complete time series of Sentinel-2 imagery for Portugal. The goal of ISA_DGT_INCD_DL_pipeline is to produce near-real-time national maps of vegetation loss for areas of forest, bushes or shrubs. Details on the pipeline are described in Deliverable E.4.4.

This deliverable describes the output of the pipeline, i.e. the vegetation loss maps in Geopackage format, its features and its attributes.

There are two data sets to be combined:

1. DGT_mask: DGT vegetation loss mask in vector format for potential areas (forest, bushes or shrubs) where vegetation loss can occur. It covers approximately 50'000 km2 of the Portuguese territory. For more information and access to data, see https://github.com/S2change/vegetation_loss/tree/main/data_info/DGT_vegetation_loss_mask
2. DL_MBPV_v3: (*mapa bimestral de perdas de vegetação v3-T5A10*). This is the final output of the ISA_DGT_INCD_DL_pipeline.
The causal agent can be “Cuts” or “Fires”. Vegetation loss maps contain both classes, in addition to the background (“no change” at the analyzed date).
The map is created for the totality of the Portuguese territory, including the parts outside the DGT_mask. The user should discard the parts not contained in DGT_mask. For more information and access to data, see https://github.com/S2change/vegetation_loss/tree/main/data_info/vegetation_loss_products

Dates and tiles

DL_MBPV_v3 is created at the tile level (see Deliverable E.4.4) for the tiles listed in Figure 1. Since the list of analyzed dates for each tile might differ (see Deliverable E4.1: User's Manual), the intersection region between tiles might contain similar vegetation loss features for close sequence of dates.

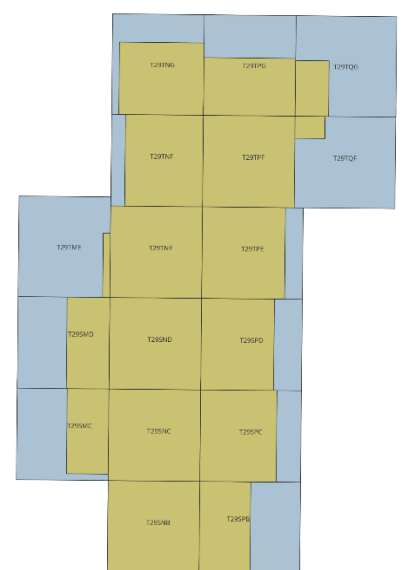


Figure 1: Sentinel-2 tiles for Portugal and minimal bounding boxes for each tile.

Since Sentinel-2 available dates have an irregular temporal distribution due the irregularity of the orbits combined with cloud cover, the user needs to decide which potential dates of change are to be analyzed for each tile. In particular, it is expectable that more potential dates are analyzed in regions with lower cloud cover and therefore denser image time series.

The choice of dates is also related to the time span of the analysis.

1. First scenario: the user might want to analyze a full year with, say, approximately 100 available Sentinel-2 daily images. To make the analysis more robust and faster, the user decides to group all dates that are separated by at most 10 days, in temporal clusters that span at most 15 days. This is illustrated in Figure 2.
2. Scenario 2: The user wants to analyze a shorter time span of, for instance, two summer months with many Sentinel-2 clear acquisition dates. To do a detailed temporal analysis, the user aggregates all dates that are separated by at most 3 days, in temporal clusters that span at most 5 days. This is illustrated in Figure 3.

The inputs that control those choices are MAX_THETA and MAX_CLUSTER_AMPLITUDE as described in Deliverable E4.1: User's Manual.

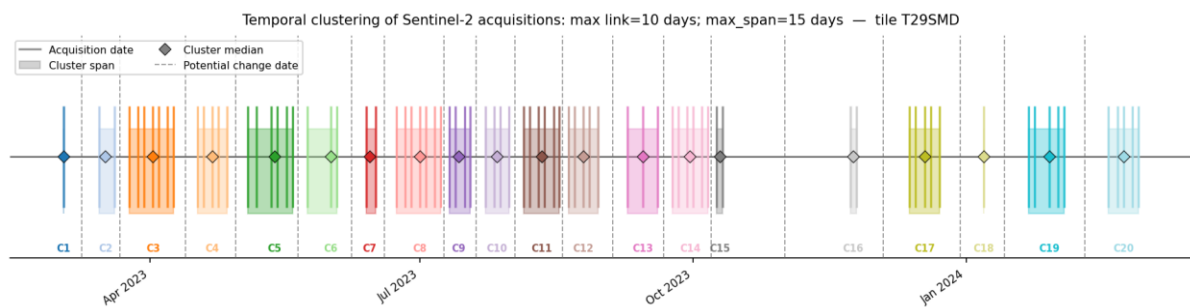


Figure 2: Temporal aggregation of dates separated by at most 10 days, in temporal clusters that span at most 15 days (parameters MAX_THETA=10; MAX_CLUSTER_AMPLITUDE=15). Colors represent different clusters of dates. Vertical colored lines represent daily Sentinel-2 observations. Dotted black vertical lines indicate potential vegetation loss dates to be analyzed.

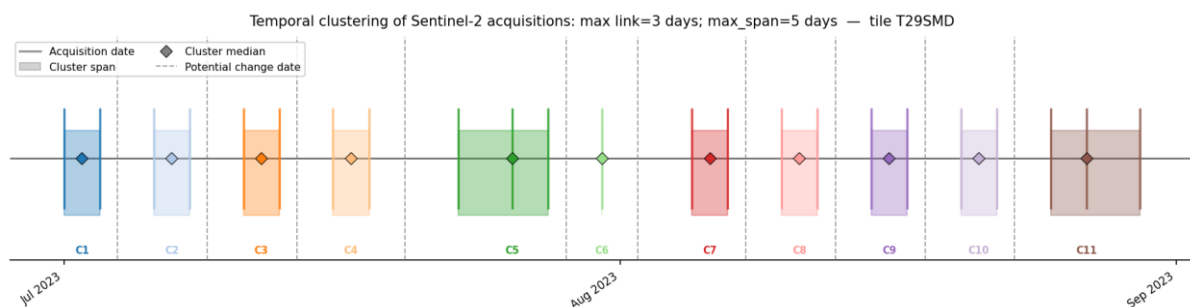


Figure 3: Temporal aggregation of dates separated by at most 3 days, in temporal clusters that span at most 5 days (parameters MAX_THETA=3; MAX_CLUSTER_AMPLITUDE=5). Colors represent different clusters of dates. Vertical colored lines represent daily Sentinel-2 observations. Dotted black vertical lines indicate potential vegetation loss dates to be analyzed.

Vegetation loss map features and attributes

Vegetation Loss maps are produced for the whole territory, in vector format, where each feature has area equal or larger to 0.5 hectares (i.e. at least 50 Sentinel-2 pixels).

Figure 4 show an example of a “Cuts” prediction and Figure 5 show also an example of adjacent “Cuts” with distinct dates. Figure 6 illustrates the detection of “Fires”.



Figure 4: Example of a (vector) feature with a detection of “Cuts” (label=1) at location Long=-8.90789, Lat=39.65848. The background Google Imagery is from 2025-2026.

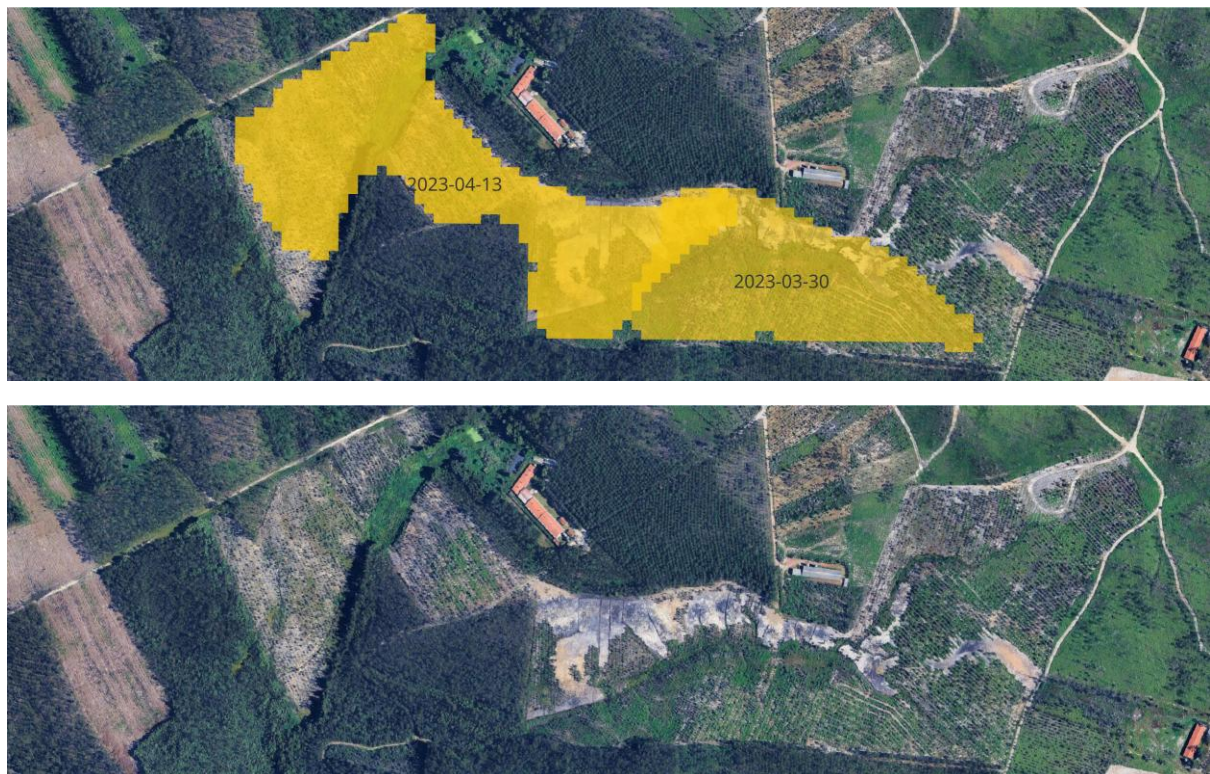


Figure 5: Adjacent detections in tile T29TNE. The background Google Imagery is from 2025-2026.



Figure 6: Example of a (vector) feature with a detection of “Fires” (label=2) at location Long= -7.61088 , Lat=40.23848. The background Google Imagery is from 2025-2026.

On the areas where tiles overlap, the range of dates is larger, and the same change event can be identified at distinct dates, as illustrated in Figure 7.

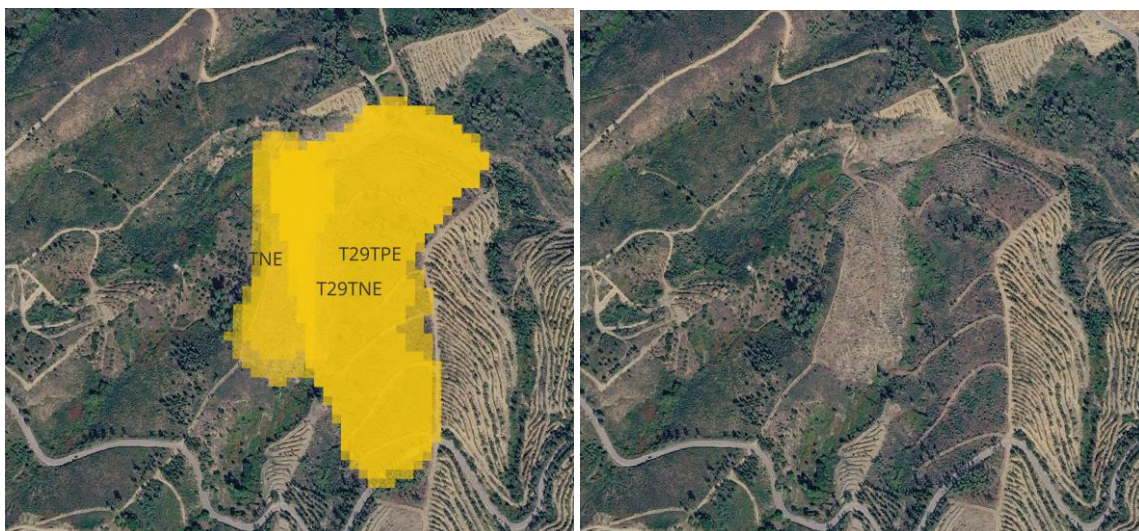


Figure 7: Detections of “Cuts” on the intersection of tiles T29TNE and T29TPE. The background Google Imagery is from 2025-2026 (Long= -7.79619, Lat=39.96118)

The attributes available for each feature are described in https://github.com/S2change/vegetation_loss/tree/main/data_info/vegetation_loss_products and include the detection date and the label (1 for “Cuts” and 2 for “Fires”).